

Classical Mechanics

Final Exam

January 11th, 2023

Exercise 1 (2 points)

John and Mary leave from a common point (with their clocks both reading zero) and travel in opposite directions with relative (and relativistic) speed v . When the time is T in Mary's clock, she sends out a light signal towards John. When John receives the signal, what is the time in his (John's) clock?

Exercise 2 (1 point)

Consider a system with N degrees of freedom under the potential $V(\mathbf{q})$; the kinetic energy T of the system is quadratic in the velocities and is given by

$$T = \sum_{i=1}^N \frac{f_i(\mathbf{q})}{2} \dot{q}_i^2.$$

Write the Lagrangian of the system, obtain the Hamiltonian H as the Legendre transform of the Lagrangian, and prove that $H = T + V$, that is, that H is the total energy of the system independently of the specific form of the functions $f_i(\mathbf{q})$.

Exercise 3 (2 points)

A particle with mass m moves along the x axis under a potential $V = Ax \sin \omega t$.

- Write the Lagrangian of the system and obtain the equation of motion for x .
- Write the Hamiltonian of the system and obtain the equations of motion for x and p_x . Prove that they are equivalent to the equation of motion you obtained from the Lagrangian formalism.
- Using the equation of motion for p_x obtained with the Hamiltonian formalism, prove that $f = p_x - \frac{A}{\omega} \cos \omega t$ is a constant of motion.
- Prove again, this time using Poisson brackets, that $f = p_x - \frac{A}{\omega} \cos \omega t$ is a constant of motion.

Exercise 4 (3 points)

Consider the following transformation for a system with two degrees of freedom

$$Q_1 = -p_1 \quad , \quad P_1 = q_1 - q_2 \quad , \quad Q_2 = q_2^2(p_1 + p_2) \quad , \quad P_2 = \frac{1}{q_2}.$$

- a) Prove that the transformation is canonical.
- b) Obtain a generating function of type $F_1(q_1, q_2, Q_1, Q_2)$ for this transformation.

Now, consider a system whose Hamiltonian is

$$H = q_2^4(p_1 + p_2)^2 + (q_1 - q_2)^2.$$

- c) Using the canonical transformation above, obtain the transformed Hamiltonian

$$\tilde{H}(Q_1, Q_2, P_1, P_2)$$

for the system.

- d) Get Hamilton's equations of motion for the transformed coordinates Q_1, Q_2, P_1 and P_2 .
- e) Integrate the equations of motion to get the trajectories $Q_1(t), Q_2(t), P_1(t)$ and $P_2(t)$. Leave the solution in terms of as many integration constants as needed.

Exercise 5 (2 points)

Consider a 1-dimensional system consisting of two masses m_1 and m_2 , which interact with each other through a central potential $V(|r_1 - r_2|)$; r_1 and r_2 are the positions of m_1 and m_2 , respectively.

- a) Write the Lagrangian $\mathcal{L}(r_1, r_2, \dot{r}_1, \dot{r}_2)$ of the system and prove that it can be rewritten as

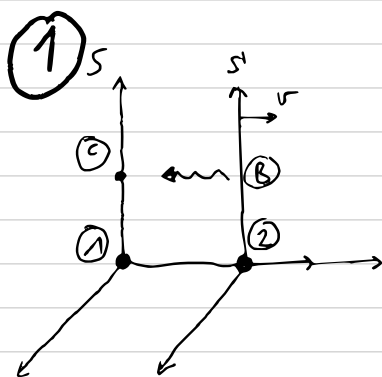
$$\mathcal{L} = \frac{M}{2}\dot{R}^2 + \frac{\mu}{2}\dot{r}^2 - V(|r|),$$

under the change of coordinates

$$R = \frac{m_1 r_1 + m_2 r_2}{M} \quad , \quad r = r_1 - r_2$$

with $M = m_1 + m_2$ and $\mu = m_1 m_2 / M$.

- b) Identify the cyclic variable in this Lagrangian, and prove that its conjugate momentum is conserved.



Event	ref. frame	
	S	S'
① A	$x_A = 0$ $t_A = 0$	$x'_A = 0$ $t'_A = 0$
② B	$t_B = ?$ $x_B = ?$	$t'_B = T$ $x'_B = 0$ 0.5 points
③ C	$t_C = ?$	

$$t_B = \gamma (t'_B + v x'_B) = \gamma T$$

$$x_B = \gamma (x'_B + v t'_B) = \gamma v T$$

0.5 points

Now, from S, the time between ② and ③ is:

$$t_{BC} = \frac{x_B}{c} = \frac{x_B}{c=1} = x_B = \gamma v T$$

Therefore, $t_C = t_B + t_{BC} = \gamma T + \gamma v T = \gamma T (1 + v) =$

$$= T \frac{(1+v)}{\sqrt{1-v^2}} = T \frac{1+v}{\sqrt{(1+v)(1-v)}} = T \sqrt{\frac{1+v}{1-v}}$$

1 point

2

$$T = \sum_i \frac{f_i(\vec{q})}{2} \dot{q}_i^2 \quad ; \quad V = V(\vec{q})$$

Lagrangian

$$\mathcal{L} = T - V = \sum_i \frac{f_i(\vec{q})}{2} \dot{q}_i^2 - V(\vec{q}) \quad \boxed{0.25 \text{ points}}$$

Hamiltonian

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i} = f_i(\vec{q}) \dot{q}_i$$

$\boxed{0.25 \text{ points}}$

$$H = \sum_i \dot{q}_i p_i - \mathcal{L} = \sum_i f_i(\vec{q}) \dot{q}_i^2 - \sum_i \frac{f_i(\vec{q})}{2} \dot{q}_i^2 + V$$

$$= \sum_i \frac{f_i(\vec{q})}{2} \dot{q}_i^2 + V(\vec{q}) = T + V \quad \square.$$

$\boxed{0.5 \text{ points}}$

3

$$V = A x \sin \omega t$$

0.5 a) $\mathcal{L} = \frac{m}{2} \dot{x}^2 - A x \sin \omega t$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = \frac{d}{dt} (m \dot{x}) = m \ddot{x} \quad \left\{ \begin{array}{l} m \ddot{x} = -A \sin \omega t \end{array} \right.$$

$$\frac{\partial \mathcal{L}}{\partial x} = -A \sin \omega t$$

0.5 b) $H = \frac{m}{2} \dot{x}^2 + A x \sin \omega t = \frac{p^2}{2m} + A x \sin \omega t$

$\uparrow$
 $p = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m \dot{x}$

$$\dot{x} = \frac{\partial H}{\partial p} = \frac{p}{m} \longrightarrow \dot{p} = m \ddot{x}$$

$$\dot{p} = -\frac{\partial H}{\partial x} = -A \sin \omega t \implies m \ddot{x} = -A \sin \omega t$$

0.5 c) $f = p - \frac{A}{\omega} \cos \omega t = ct$; $\left[\frac{df}{dt} \right] = \dot{p} + \frac{A}{\omega} \sin \omega t \overset{?}{=} 0$

0.5 d) $\frac{\partial f}{\partial t} = \frac{A \sin \omega t}{\omega} = A \sin \omega t$

$$[f, H] = \frac{\partial f}{\partial x} \frac{\partial H}{\partial p} - \frac{\partial H}{\partial x} \frac{\partial f}{\partial p} = 0 - A \sin \omega t \cdot 1 = -A \sin \omega t$$

$$\left[\frac{df}{dt} \right] = [f, H] + \frac{\partial f}{\partial t} = -A \sin \omega t + A \sin \omega t = 0 \quad \square$$

(4)

$$Q_1 = -p_1 ; P_1 = q_1 - q_2 ; Q_2 = q_2^2 (p_1 + p_2) ; P_2 = \frac{1}{q_2}$$

0.6

a) Necessary & sufficient condition $[Q_i, Q_j] = 0$, $[P_i, P_j] = 0$ and

0.1 $[Q_i, P_j] = \delta_{ij}$. Let's see:

0.1

$$[Q_1, P_1] = \frac{\partial Q_1}{\partial q_1} \frac{\partial P_1}{\partial p_1} - \frac{\partial P_1}{\partial q_1} \frac{\partial Q_1}{\partial p_1} + \frac{\partial Q_1}{\partial q_2} \frac{\partial P_1}{\partial p_2} - \frac{\partial P_1}{\partial q_2} \frac{\partial Q_1}{\partial p_2} =$$

$$= - \frac{\partial P_1}{\partial q_1} \frac{\partial Q_1}{\partial p_1} = -(-1) \cdot 1 = 1 \checkmark$$

0.1

$$[Q_2, P_2] = \frac{\partial Q_2}{\partial q_1} \frac{\partial P_2}{\partial p_1} - \frac{\partial P_2}{\partial q_1} \frac{\partial Q_2}{\partial p_1} + \frac{\partial Q_2}{\partial q_2} \frac{\partial P_2}{\partial p_2} - \frac{\partial P_2}{\partial q_2} \frac{\partial Q_2}{\partial p_2} =$$

$$= - \frac{\partial P_2}{\partial q_2} \frac{\partial Q_2}{\partial p_2} = \frac{1}{q_2^2} q_2^2 = 1 \checkmark$$

0.1

$$[Q_1, Q_2] = \frac{\partial Q_1}{\partial q_1} \frac{\partial Q_2}{\partial p_1} - \frac{\partial Q_2}{\partial q_1} \frac{\partial Q_1}{\partial p_1} + \frac{\partial Q_1}{\partial q_2} \frac{\partial Q_2}{\partial p_2} - \frac{\partial Q_2}{\partial q_2} \frac{\partial Q_1}{\partial p_2} =$$

$$= 0 \checkmark$$

0.1

$$[P_1, P_2] = \frac{\partial P_1}{\partial q_1} \frac{\partial P_2}{\partial p_1} - \frac{\partial P_2}{\partial q_1} \frac{\partial P_1}{\partial p_1} + \frac{\partial P_1}{\partial q_2} \frac{\partial P_2}{\partial p_2} - \frac{\partial P_2}{\partial q_2} \frac{\partial P_1}{\partial p_2} = 0 \checkmark$$

0.1

$$[Q_1, P_2] = \frac{\partial Q_1}{\partial q_1} \frac{\partial P_2}{\partial p_1} - \frac{\partial P_2}{\partial q_1} \frac{\partial Q_1}{\partial p_1} + \frac{\partial Q_1}{\partial q_2} \frac{\partial P_2}{\partial p_2} - \frac{\partial P_2}{\partial q_2} \frac{\partial Q_1}{\partial p_2} = 0 \checkmark$$

0.1

$$[Q_2, P_1] = \frac{\partial Q_2}{\partial q_1} \frac{\partial P_1}{\partial p_1} - \frac{\partial P_1}{\partial q_1} \frac{\partial Q_2}{\partial p_1} + \frac{\partial Q_2}{\partial q_2} \frac{\partial P_1}{\partial p_2} - \frac{\partial P_1}{\partial q_2} \frac{\partial Q_2}{\partial p_2} = -q_2^2 + q_2^2 = 0 \checkmark$$

b) Implicit transformation equations for F_1

0.6

$$p_i = \frac{\partial F_1}{\partial q_i} \quad ; \quad I_i = -\frac{\partial F_1}{\partial a_i}$$

We start by writing p_1, p_2, I_1 , and I_2 as functions of q_1, q_2, a_1 , and a_2 .

$$\textcircled{1} \quad p_1 = -Q_1 = \frac{\partial F_1}{\partial q_1}$$

$$\textcircled{2} \quad I_1 = q_1 - q_2 = -\frac{\partial F_1}{\partial a_1}$$

0.2P

$$\textcircled{3} \quad I_2 = \frac{1}{q_2} = -\frac{\partial F_1}{\partial a_2}$$

$$\textcircled{4} \quad p_2 = \frac{Q_2}{q_2^2} - p_1 = \frac{Q_2}{q_2^2} + Q_1 = \frac{\partial F_1}{\partial q_2}$$

Now, we integrate:

$$\begin{aligned} \textcircled{4} \Rightarrow F_1 &= \int dq_2 \left(\frac{Q_2}{q_2^2} + Q_1 \right) + f(q_1, a_1, a_2) = \\ &= -\frac{Q_2}{q_2} + Q_1 q_2 + f(q_1, a_1, a_2) \end{aligned}$$

0.2P

$$\textcircled{3} \Rightarrow F_1 = \int da_2 \frac{1}{q_2} + g(q_1, q_2, a_1) = -\frac{Q_2}{q_2} + g(q_1, q_2, a_1)$$

$$\textcircled{2} \Rightarrow F_1 = -\int da_1 (q_1 - q_2) + h(q_1, q_2, a_2) = -Q_1 q_1 + q_2 Q_1 + h$$

$$\textcircled{1} \Rightarrow F_1 = -\int dq_1 Q_1 + z(q_2, a_1, a_2) = -q_1 Q_1$$

All in all, we have that all of the above are compatible with:

$$F_1 = -\frac{Q_2}{q_2} + Q_1(q_2 - q_1)$$

0.2 p

0.6 c) By noting that $P_1 = (q_1 - q_2)$ and $Q_2 = q_2^2(p_1 + p_2)$ we have that $\tilde{H}$ is trivially

$$\tilde{H} = Q_2^2 + P_1^2$$

0.6 d) Equation of motion

$$\textcircled{1} \dot{P}_1 = -\frac{\partial \tilde{H}}{\partial Q_1} = 0$$

$$\textcircled{3} \dot{P}_2 = -\frac{\partial \tilde{H}}{\partial Q_2} = -2Q_2$$

$$\textcircled{2} \dot{Q}_1 = \frac{\partial \tilde{H}}{\partial P_1} = 2P_1$$

$$\textcircled{4} \dot{Q}_2 = \frac{\partial \tilde{H}}{\partial P_2} = 0$$

0.6 e) $\textcircled{1} \Rightarrow [P_1 = \alpha_1] \text{ (constant)}$

$$\textcircled{2} \Rightarrow \dot{Q}_1 = 2\alpha_1 \Rightarrow [Q_1 = 2\alpha_1 t + \beta_1]$$

$$\textcircled{4} \Rightarrow [Q_2 = \alpha_2] \text{ (constant)}$$

$$\textcircled{3} \Rightarrow \dot{P}_2 = -2\alpha_2 \Rightarrow [P_2 = -2\alpha_2 t + \beta_2]$$

5

Done in class and in lecture notes